

Majorana zero modes in silicon: quantified constraints on the conduction band and a conditional hole-based route

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Abstract

We present a computational feasibility and falsification study of Majorana zero modes in silicon nanowires (Bogoliubov–de Gennes, valley-resolved, six-band $k \cdot p$ + Poisson; every quoted number regenerates from the public repository and is machine-verified against its numerical ledger). *Conduction band*: three quantified obstructions — a spin–orbit shortfall, the single-channel $g \approx 2$ Zeeman ceiling, and valley physics, where we prove smooth interfaces benign under inter-valley pairing but find each atomic step acts as a near- π Josephson junction ($\Delta\varphi \approx 0.85\pi$) that caps the gap at $\sim 18\%$ from the first step. With miscut-derived step ensembles, device *yield* is the binding statistic: 2–5% for unaligned wires even on premium wafers, 50–68% for wires within 10° of step edges. Within this model class, conduction-band Si is strongly disfavored unless step morphology is specifically controlled. *Valence band*: holes evade valley physics; our six-band [110]-channel treatment (the orientation of every benchmark device) resolves the long-standing model-vs-experiment g discrepancy as predominantly an axes artifact and yields gate-stable $g_{x'} = 1.4$ – 1.7 , $m^* \approx 0.20$, α up to $0.075 \text{ eV \AA}$. Adjudicated against explicit operational tiers with the dynamic tunneling self-energy as the central estimate (static values are upper bounds): measured thick Si:B reaches 8 – $10 \mu\text{eV}$ at tilted field (marginal detection tier); hypothetical[†] Pauli-limited thin-film Si:B, $\sim 13 \mu\text{eV}$; hypothetical[†] hard-gap Al/Si-hole, 17 – $18 \mu\text{eV}$ (the only scenario approaching the nonlocality tier and surviving quasiparticle poisoning at realistic temperatures). The binding materials constraint is correlated electrostatic disorder: calibrated against oxide-interface charge, standard trap densities produce potential disorder orders of magnitude beyond tolerance, so interface quality — not step morphology or parent choice — gates any MOS-style silicon implementation. Two measurements would do the most to settle the platform: the parallel critical field of thin Si:B films and the device-geometry [110] hole g -tensor. Qubit relevance is not established.

1 Introduction

Semiconductor–superconductor hybrid nanowires remain the most studied route to MZMs [1, 2, 3], and a decade of experiments—from the first InSb signatures [4] through the disorder-driven retractions and reappraisals [5] to protocol-based standards [6]—has made one lesson clear: material quality and honest accounting of trivial explanations dominate the problem. Silicon, the most controllable semiconductor in existence, has been conspicuously absent from this program, for reasons usually stated qualitatively: weak spin–orbit coupling (SOC), a small g -factor, and conduction-band valley degeneracy.

This paper is a computational feasibility and falsification study: it makes those reasons quantitative, finds that one of them hides a qualitatively new failure mechanism, and shows that silicon

holes evade it — while facing their own multiband g -tensor, electrostatic, and parent-interface constraints, quantified below. Our contributions: (i) a quantified electron verdict (Secs. 3.1, 3.2); (ii) an exact unitary equivalence making smooth-interface valley splitting benign under the physical inter-valley pairing channel, and the identification of single-atomic interface steps as near- π Josephson junctions that sharply degrade protection from the first step (single-step cap $\sim 18\%$) (Sec. 3.3); (iii) a Luttinger–Kohn (LK) derivation of the gate-field covariation of hole parameters, which erodes free-parameter optimism but preserves a viable design (Sec. 4); and (iv) orientation- resolved feasibility maps for two silicon-compatible parents—laser-doped superconducting Si:B [20, 21, 23, 24] and Al films—including thin-film design targets, metallization renormalization, and quasiparticle-poisoning limits (Sec. 4.2). Throughout, spectral diagnostics are cross-checked by recursive-Green’s-function transport (Sec. 5).

2 Model and methods

We use the standard single-channel BdG Hamiltonian

$$H = \left(\frac{\hbar^2 k^2}{2m^*} - \mu + V(x) \right) \sigma_0 + \alpha k \sigma_z + E_Z \sigma_x \quad (\text{normal part}), \quad \Delta i \sigma_y \quad (\text{pairing}), \quad (1)$$

discretized with exact particle–hole symmetry by construction, topological for $E_Z^2 > \Delta^2 + \mu^2$. Validation (repository Fig. 1): gap closing at the analytic boundary; end-localized zero modes; exponential length splitting; finite-wire vs analytic bulk-gap agreement. Valley physics uses an $8N$ basis with *inter-valley* singlet pairing $\Delta(\nu_x \otimes i\sigma_y)$ —the physical zero-momentum channel since the relevant valleys are time-reversed partners—with arbitrary complex valley-orbit (VO) profiles $\lambda(x)$ and a gauge-covariant phase-locked Dresselhaus-type SOC option. Hole parameters derive from a four-band LK model of the fin cross-section (validated against exact [100] bulk masses) with vertical field, bare (κ) and orbital Zeeman terms. Transport uses a dimension-generic RGF with Ando wave-matched normal leads and an $E = 0$ reflection invariant $\text{sign det } r$ (kwant-equivalent; validated against closed-chain spectra to 10^{-13}). A test suite pins every exact claim. Δ is a *static* induced gap (the $\omega = 0$ limit of the standard tunneling self-energy); parent-gap field suppression enters via $\Delta_{ind} \leq \Delta_p(B)$ and metallization via the quasiparticle weight Z [7, 25, 26] (Supplement S6); the full dynamical self-energy is solved as a control in Sec. 6 and validates the static- Z treatment directionally while lowering its numbers by 24–30%. All caricature choices (GL/Zee-man-linear parent gaps, iid onsite disorder with correlation length dx , hard-wall LK geometry) are stated where used; complete per-figure parameter tables, seed families, and discretization/length/seed/grid convergence studies are in Supplement S4–S5. Throughout, claims carry one of three epistemic levels, stated where ambiguity is possible: *exact within model* (the valley equivalence theorem, the $g = 2$ ceiling, topological criteria), *numerical within model* (gap optimizations, step ensembles, disorder thresholds, transport), and *device extrapolation* (parent feasibility, interface assumptions, fabrication design rules).

2.1 Operational criteria

“Usable” is not a single threshold; we therefore evaluate every platform scenario of Table 1 against three named, quantitative tiers.

Tier S (spectroscopy/detection). Topological gap $\Delta_{\text{top}} \geq 2k_B T_{\text{eff}}$ at $T_{\text{eff}} = 50 \text{ mK}$, i.e. $\Delta_{\text{top}} \geq 8.6 \mu\text{eV}$ ($k_B \times 50 \text{ mK} = 4.3 \mu\text{eV}$), and Majorana splitting $E_0 \leq 0.1 \mu\text{eV}$ at $L \leq 3 \mu\text{m}$. The

gap threshold ensures the subgap feature is resolvable against thermal smearing in tunneling spectroscopy at a routinely achieved electron temperature; the splitting threshold keeps the zero-bias peak unsplit at spectroscopic resolution and is met by our clean finite-wire checks ($E_0 = 3 \times 10^{-4} \mu\text{eV}$ at $2 \mu\text{m}$, repository Fig. 1; hole-wire check $E_0 \approx 0$ at $3 \mu\text{m}$, `fig8.hole_wire_finite_check`).

Tier N (nonlocality/fusion). Additionally $\Delta_{\text{top}} \geq 20 \mu\text{eV}$ and disorder half-gap threshold $W_{1/2} \geq 200 \mu\text{eV}$. The $20 \mu\text{eV}$ value is the protection scale used throughout this paper (e.g. the α requirement of Sec. 3.1, `fig3.alpha_needed_for_20ueV`), giving $\gtrsim 4 k_B T$ headroom for braiding/fusion protocols at 50 mK; the disorder threshold demands a finite margin against realistic potential fluctuations, against which the transport-level $W_{1/2} \approx 548 \mu\text{eV}$ of the engineered-electron case (Sec. 5) and the hole-wire gap persisting to $W \approx 300 \mu\text{eV}$ (`fig8.hole_wire_finite_check`) are measured.

Tier Q (parity-coherent qubit). Additionally a sustainable parent quasiparticle fraction $x_{qp} \leq 10^{-6}$ at $T_{\text{eff}} \geq 50 \text{ mK}$ given the parent gap *at the operating field*: 10^{-6} is the standard parity-lifetime requirement, and 50 mK is the coldest electron temperature routinely demonstrated for quasiparticle thermalization. From `qp_poisoning.py` (Supplement S7): the Si:B operating points ($\Delta_p(B^*) = 29\text{--}33 \mu\text{eV}$) require $T_{\text{eff}} \leq 25\text{--}29 \text{ mK}$ and therefore fail; the Al reference ($\Delta_p = 150 \mu\text{eV}$ at 1 T) requires only $T_{\text{eff}} \leq 130 \text{ mK}$ and passes.

3 Conduction band: three independent obstructions

3.1 Spin-orbit shortfall

Maximizing the topological gap over $(\mu, B \leq 2 \text{ T})$ at measured intrinsic Rashba strengths gives $0.16 \mu\text{eV}$ (typical, $\alpha = 10^{-4} \text{ eV \AA}$) to $1.5 \mu\text{eV}$ (optimistic, 10^{-3}): below dilution-fridge protection scales and removed by onsite disorder far weaker than realistic interface fluctuations. Reaching a robust $20 \mu\text{eV}$ requires $\alpha \approx 0.015$ (idealized) to $0.021 \text{ eV \AA}$ (with parent-gap suppression $\Delta(B)$)—an optimistic lower bound given unmodeled orbital and metallization corrections. Micromagnet arrays [8, 9] reach this scale in principle; a 2025 InAs experiment measured synthetic $\alpha = 0.022 \text{ eV nm} = 0.22 \text{ eV \AA}$ [10]—an order of magnitude *above* our silicon requirement, though demonstrated in a lighter-mass material with a mature hybrid stack, and with micromagnet fringe-field disorder unmodeled here. “Engineered-electron silicon” is therefore marginal rather than absurd—but the valley physics below stacks on top of it.

3.2 The $g = 2$ catch-22

Since $E_Z(2 \text{ T}) = 116 \mu\text{eV}$ for $g = 2$, an induced gap $\Delta \geq 116 \mu\text{eV}$ cannot reach the topological phase for $B \leq 2 \text{ T}$ within the single-channel $g = 2$ model; in the SOC-limited regime relevant to engineered devices, increasing interface transparency *reduces* the achievable gap, with an optimal parent gap $\Delta_0 \approx 98 \mu\text{eV}$ once $\Delta(B)$ suppression is included. This sharpens, for $g = 2$, the weak-coupling optimum identified by Cole, Das Sarma and Stanescu [7]. Pauli-limited parents satisfy the constraint automatically (Sec. 4.2).

3.3 Valley physics: an exact equivalence and step junctions

Equivalence theorem. With inter-valley pairing and *any* uniform VO phase, the wire maps unitarily onto two decoupled bands at $\mu \mp |\lambda|$ with pairing $\pm \Delta$ (proof in Supplement S2; the key

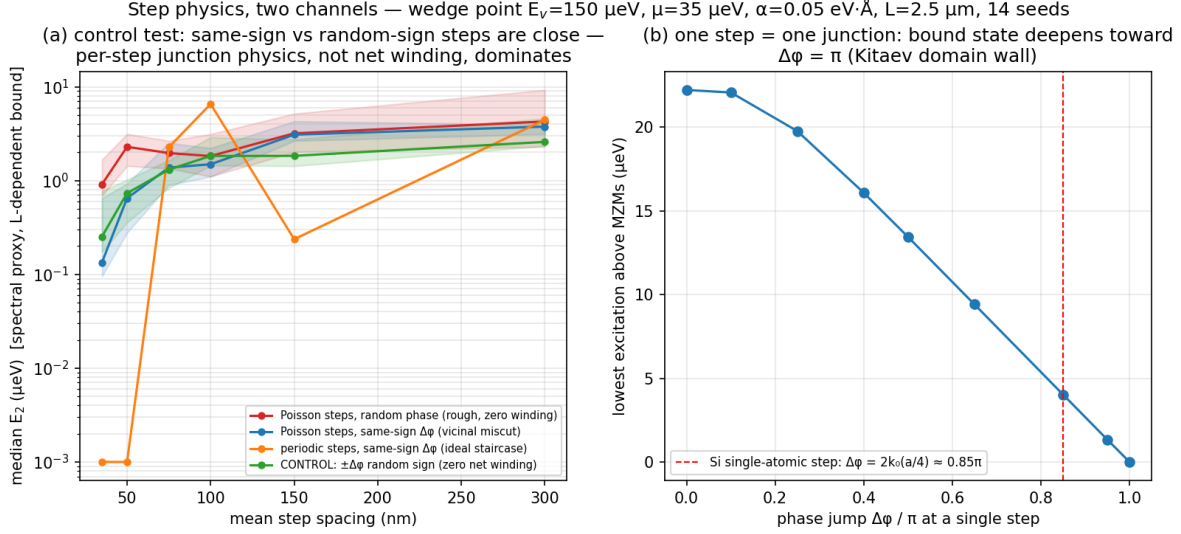


Figure 1: Step physics in the inter-valley-paired electron wire. (a) Controlled scenario comparison at fixed jump magnitude: same-sign (vicinal) and sign-randomized (zero-winding control) staircases are statistically indistinguishable—per-step junction physics, not winding, dominates. (b) A single VO phase step is a Josephson-like junction; the Si single-step value 0.85π sits near the Kitaev domain-wall point π .

step is the exact invariance of the inter-valley pairing under the valley rotation, $U_\phi D U_\phi^T = D$; test-suite tolerance $10^{-6} \mu\text{eV}$ on every eigenvalue, spot checks $3 \times 10^{-10} \mu\text{eV}$, for valley-scalar and phase-locked Dresselhaus SOC alike): smooth interfaces are benign, and valley splitting $\gtrsim 100 \mu\text{eV}$ buys single-valley topology, as the simple wedge picture suggests.

Step junctions. Real interfaces are stepped. That steps shift the VO phase by $2k_0\delta z$ is established quantum-dot physics [12, 11]; the new statement here is its consequence under inter-valley *pairing*: projecting the pairing onto the local valley bands gives each band an order parameter of phase $-\phi(x)$ (Supplement S3, exact), so each single-atomic step, $\Delta\phi = 2k_0(a/4) \approx 0.85\pi$, is a Josephson junction accidentally close to π in silicon. The bound-state depth is determined numerically; the transparent short-junction formula $\Delta_t |\cos(\Delta\phi/2)|$ tracks it to 10–25% (Supplement Table 1). A controlled comparison (same-sign staircase vs sign-randomized steps of equal magnitude) shows net phase *winding* is irrelevant at realistic densities; the damage is per-step junction physics: a single step binds a subgap state at $4.0 \mu\text{eV}$ (18% of the $22 \mu\text{eV}$ clean gap), deepening to a Kitaev domain-wall zero mode at $\Delta\phi = \pi$ [Fig. 1(b)]. Ensembles at 50 nm step spacing retain only ~ 0.7 – $2.3 \mu\text{eV}$ (median, L -dependent upper bounds). Bunched double-height steps ($\Delta\phi = 1.7\pi \equiv -0.3\pi$) are $\sim 2\times$ milder—step-bunched vicinal surfaces are preferable—while VO-amplitude suppression near steps [12] worsens the damage; wires laid along step edges avoid it entirely ($|\cos \chi|$ scaling). Smooth phase gradients are a real but subdominant second channel, suppressed at large valley splitting. Transport (Sec. 5) adds a warning: step-dense wires become Anderson insulators whose end-to-end transport gap *grows* with length ($0.5/5.6/23 \mu\text{eV}$ at $L = 1.5/3/6 \mu\text{m}$) while localized subgap states persist—a hard-looking transport gap does not certify protection. The falsifiable consequence: induced-gap hardness in proximitized Si *electron* devices should degrade with miscut/step density from the very first step, testable by gap spectroscopy on standard miscut wafer series; holes, having no conduction-band valleys, are not subject to this particular mechanism (they remain subject to every other interface imperfection).

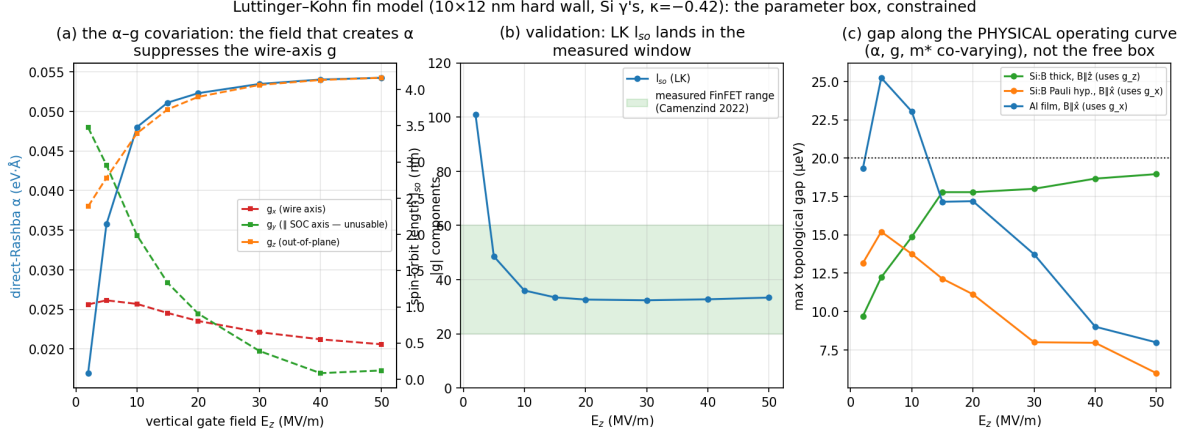


Figure 2: Luttinger–Kohn constraint on the hole platform: (a) α – g covariation under the gate field; (b) validation of l_{so} against FinFET measurements; (c) accessible gap along the physical operating curve for three parent/orientation scenarios.

4 Valence band

4.1 Six-band $[110]$ $k \cdot p$ + Poisson: the primary hole model

Quasi-1D hole channels in Si FinFETs have measured strong, gate-tunable SOC [15, 19, 14] and anisotropic g -tensors [16] — and every benchmark device is a $[110]$ -oriented channel on a (001) surface. Our primary hole model is therefore the six-band ($\Gamma_8 \oplus \Gamma_7$) $k \cdot p$ fin Hamiltonian rotated to the $[110]$ channel, with self-consistent tri-gate electrostatics available as a control (Supplement S10; validated against bulk dispersions, the four-band limit to 0.006%, and Kramers/Hermiticity to machine precision). It gives a gate-stable wire-axis $g_{x'} = 1.43$ – 1.66 across all benchmark geometries and the full field range, $m^* \approx 0.20$, α up to $0.075 \text{ eV \AA}$ ($l_{so} \approx 50 \text{ nm}$), and $g_{y'} = 2.0$ – 2.2 inside the measured window — within 0.3–0.6 of the measured along-fin $g \approx 1.9$ – 2.3 , a residual of the size and sign expected from electrostatic asymmetry (which raises $g_{x'}$ and α together, Supplement S10.2) and 0.1–0.2% process strain [17].

Pedagogical $[100]$ limit and the covariation caution. The four-band hard-wall $[100]$ model that earlier drafts used as primary reproduces $l_{so} = 32$ – 101 nm but illustrates a trap the $[110]$ geometry largely avoids [Fig. 2]: there, the vertical field that builds α (to $0.054 \text{ eV \AA}$) suppresses the wire-axis g_x from 1.1 to 0.5 while raising g_z to 2.4–4.2 ($m^* \approx 0.44$, SOC axis in-plane transverse) — a harmful α – g covariation. Its seven-geometry bracket keeps $g_x \in [0.4, 1.1]$, an apparent factor-of-two below measurement that the six-band $[110]$ treatment resolves as predominantly an axes artifact: the split-off band moves g_x down (consistent with Ref. [17]: its envelope weight is negligible), electrostatics is a $\pm 25\%$ geometry-signed correction, and the $[110]$ rotation closes $\sim 60\%$ of the gap. All platform numbers below use the $[110]$ tensors; the frozen $[100]$ /static values are retained in Supplement S10.3 for provenance.

4.2 Parents, orientation, and the surviving designs

Field-orientation maps [Fig. 3], using the tilted-field topological criterion of Osca, Ruiz and Serra [13], under both g -tensors and two parents give the design rules: (i) an **Al film** parent is locked to in-plane fields but strong— $34 \mu\text{eV}$ metallization-renormalized at the center point, up to $55 \mu\text{eV}$ bare over orientations (Fig. 3, not renormalized)—if a hard-gap Al/Si-hole interface can be fab-

Table 1: Scenario adjudication under the operational tiers (Sec. 2.1), six-band [110] parameters, dynamic self-energy as the central estimate (static values are upper bounds). [†]=rests on an undemonstrated materials property. All hole rows additionally inherit the binding oxide-charge constraint of Supplement S11.4.

scenario	demonstrated?	unproven requirement	gap (μeV)			tier verdict	main failure mode	decisive measurement
			bare	renorm.	dyn.			
Si e^- intrinsic	partial (interface)	—	1.6	—	—	FAIL	SOC shortfall	—
Si e^- engineered [†]	no	$\alpha \approx 0.05 \text{ eV \AA}$ in Si; step control	29.8	—	—	COND	step yield (2–5% unaligned; 50–68% at $\chi \leq 10^\circ$)	miscut gap spectroscopy
Si:B thick (measured), tilted	parent: yes (bulk)	none (measured inputs)	12.0–17.1	10.3–12.6	8.4–10.5	S (marginal)	QP poisoning ($T_{\text{eff}} \leq 25\text{--}29 \text{ mK}$); correlated $\mu(x)$	device-geometry [110] g -tensor
Si:B thin film, Pauli [†]	no	B_d at Pauli limit; $T_c(d)$	21–23	16.6–18.7	12.5–13.0	S [†] (cond.)	film hypothesis; QP ($\leq 29 \text{ mK}$)	thin-film B_d
Al film [†]	no	hard-gap Al/Si-hole interface	36.6–40.5	26.0–32.1	16.9–18.4	S [†] , near-N [†] , Q [†] (130 mK)	interface nonexistent; correlated $\mu(x) \lesssim 25\text{--}50 \mu\text{eV RMS}$	interface demonstration

ricated (none exists); (ii) a **thick Si:B** parent [20, 21, 22, 23] is orientation-tolerant and, using only measured critical fields ($B_{c2} \approx 0.1\text{--}0.4 \text{ T}$, orbital-limited), supports $10\text{--}19 \mu\text{eV}$ at fields tilted toward out-of-plane (riding g_z ; the center point is $11.0 \mu\text{eV}$ bare / $9.8\text{--}10.1$ renormalized on finer optimizer grids, Supplement S5, moves over $8.0\text{--}10.6 \mu\text{eV}$ under $\pm 20\%$ parent-model variations, and falls to $7.6 \mu\text{eV}$ under the dynamic self-energy control of Sec. 6); the in-plane wire-axis direction is g -starved ($3\text{--}9 \mu\text{eV}$). (iii) The **thin-film Si:B** route turns the “Pauli-limited” hypothesis into a fabrication target: with coherence lengths from measured perpendicular critical fields ($\xi = 29\text{--}57 \text{ nm}$), films of $d \lesssim 18\text{--}36 \text{ nm}$ have parallel orbital critical fields above the Pauli limit $B_P = 1.11 \text{ T}$, and $d = 10\text{--}20 \text{ nm}$ films support **$21\text{--}27 \mu\text{eV}$** (modulo unmodeled $T_c(d)$ suppression). Quasiparticle poisoning bounds the role of the measured-field stack: at operating fields the Si:B parent gap is $29\text{--}33 \mu\text{eV}$, so $x_{qp} = 10^{-6}$ requires $T_{\text{eff}} \leq 25\text{--}29 \text{ mK}$ (Al reference: 130 mK)—a detection platform, not a parity-coherent qubit, absent parent-gap engineering. Table 1 adjudicates the surviving scenarios at the [110] parameters with the dynamic self-energy as the central estimate. The [110] geometry changes the design logic: with $g_{x'} \approx 1.6$ usable *in-plane along the wire*, film parents no longer need the orientation compromise, while the measured-field Si:B stack is strongest at tilted field — $8.4\text{--}10.5 \mu\text{eV}$ dynamic (Supplement S11.1; the static $12.5\text{--}17.1 \mu\text{eV}$ values are superseded) — and only marginal in-plane ($3.8\text{--}5.3 \mu\text{eV}$). Tuning tolerances are not the obstacle: the dynamic optima tolerate $\pm 14\text{--}30 \mu\text{eV}$ in μ , $\gtrsim 35\%$ in B , and a factor ≥ 3 in Γ (Supplement S11.1). Transport-level validation passes at the hole operating points exactly as for electrons, 50 mK tunneling spectroscopy brackets the true gap to $\pm 35\%$, and a failure-mode library shows the discriminating diagnostic pair is the scattering invariant co-read with the transport gap (Supplement S11.2). Multi-subband occupancy and spatially fluctuating coupling $\Gamma(x)$ are both benign (Supplement S11.5). The decisive negative result is the charge-trap calibration (Supplement S11.4): oxide-interface charge at standard densities ($10^{10}\text{--}10^{11} \text{ cm}^{-2}$) produces correlated potential disorder orders of magnitude beyond the $\sim 25\text{--}50 \mu\text{eV RMS}$ tolerance, so *interface quality is the gating materials requirement* for any MOS-style implementation — favoring remote-doping-free Si/SiGe-style channels, dramatically better oxides, or metallic-density screening.

5 Transport-level verification

RGF transmission with normal leads validates the clean wire (transport gap 37.3 vs bulk $36.8 \mu\text{eV}$) and provides $\text{sign det } r$ as a topological invariant. Under onsite disorder the spectral proxy (third-smallest $|E|$) is *conservative* at moderate strength (localized states carry no current): the engineered-electron case’s half-gap threshold moves from $W \approx 410$ to $\approx 548 \mu\text{eV}$, and breakdown is abrupt—a class-D zero-energy transmission anomaly coinciding with invariant sign flips—rather than a soft closing [Fig. 5]. The same machinery extended to eight-orbital valley cells yields the Anderson-insulator warning of Sec. 3.3.

Field-orientation maps ($\mu=0$): star = in-plane along wire; triangle = out-of-plane.
 LK says: out-of-plane B (g_z) with a thick parent; the in-plane/wire direction is g-starved.

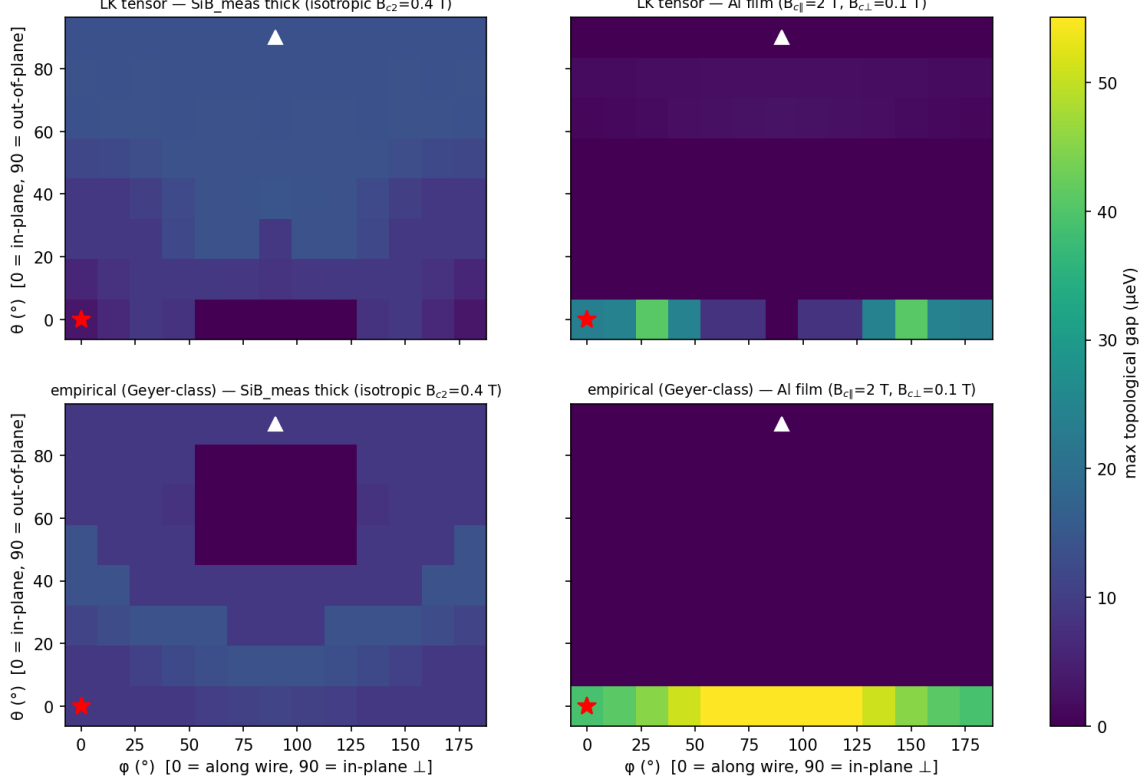


Figure 3: Maximum topological gap vs field orientation for LK (top) and empirical (bottom) hole g -tensors, with thick measured-Si:B (left) and Al-film (right) parents. Star: in-plane along the wire; triangle: out-of-plane.

6 Realism controls

Four controls test the headline numbers against the caricatures they rest on (Supplement S9). (i) *Dynamic self-energy*: replacing the static Δ with the full $\Sigma(\omega)$ and re-optimizing over tunnel coupling Γ gives center-point gaps of $7.6 \mu\text{eV}$ (measured Si:B, $\Gamma^* = 10 \mu\text{eV}$), $14.9 \mu\text{eV}$ (Pauli Si:B, $\Gamma^* = 30$), and $24.2 \mu\text{eV}$ (Al, $\Gamma^* = 45$) — $1.4\text{--}2.1\times$ below the static bare values and $24\text{--}30\%$ below the static-renormalized ones, which are hereby demoted to upper bounds; Dynes broadening adds a $\gamma_D/\Delta_p \lesssim 10^{-3}$ requirement for 50 mK operation. (ii) *Orbital coupling*: Peierls-phase controls show $< 0.05\%$ gap suppression at fin widths for the tilted Si:B design — the orientation maps survive. (iii) *Pairing-channel mixing*: intra-valley admixture η never weakens the step-junction mechanism (the step reappears as a pairing-phase junction; $2\times$ deeper bound state by $\eta^* \approx 0.24$) while healing staircase ensembles — our $\eta = 0$ ensembles are conservative, and $\eta \ll \eta^*$ is expected since intra-valley pairing requires $2k_0$ momentum transfer. (iv) *Morphology and correlated disorder*: with miscut-derived terrace ensembles, device *yield* ($25\text{--}33\%$ failure fraction at $0.05\text{--}0.1^\circ$ miscut), not the median gap, is binding; wires along step edges or step-bunched templates empty the failure tail. Correlated $\mu(x)$ disorder at $50 \mu\text{eV}$ RMS collapses the hole-platform median gap to $3.5\text{--}6.8 \mu\text{eV}$ where the iid baseline retains $29 \mu\text{eV}$: smooth electrostatics is the binding fabrication requirement, and our earlier iid thresholds overstate robustness by $4\text{--}8\times$ at fixed RMS.

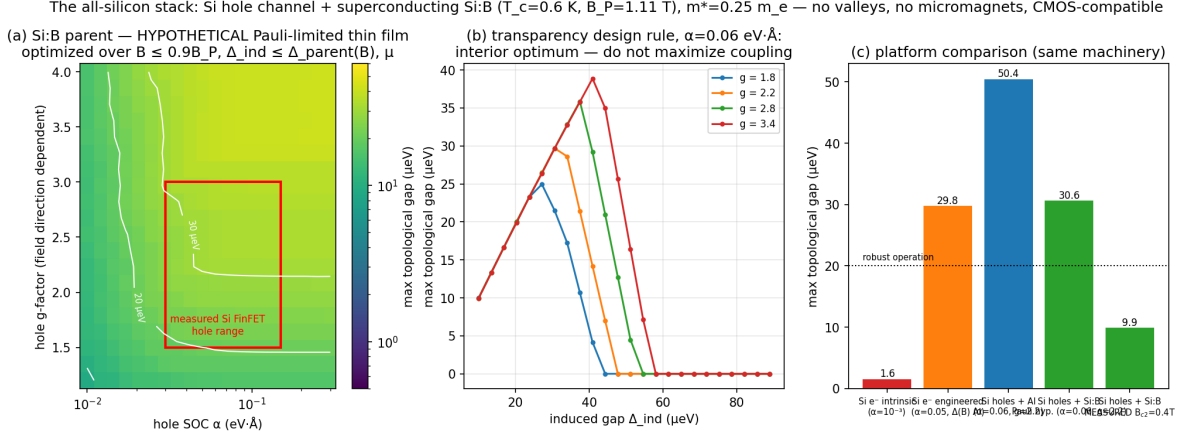


Figure 4: The all-silicon stack under the (hypothetical) Pauli-limited Si:B parent: (a) gap over the empirical (α, g) box; (b) the transparency design rule (interior optimum); (c) platform comparison including the measured-field Si:B scenario.

7 Discussion

Two measurements would do the most to settle the silicon question. First, the **parallel critical field of thin (≤ 20 nm) laser-doped Si:B films**: if it approaches the Pauli limit, as measured coherence lengths permit, the all-silicon hole stack supports $\gtrsim 20 \mu\text{eV}$ gaps in a CMOS-compatible, same-crystal materials system; if films remain orbital-limited at ~ 0.4 T, the stack is capped near $10\text{--}15 \mu\text{eV}$ and detection-only. Second, the **wire-axis hole g -factor and SOC axis** in the actual gated geometry: our LK bracket and the published FinFET values differ by $2\times$ along the one direction that matters for film parents. Both measurements are standard characterization, neither requires a working topological device, and the electron-side miscut prediction offers a third, independent test of the framework. We caution that the step-junction mechanism, the orientation maps, and all platform numbers inherit the stated caricatures (single channel, hard-wall cross sections, parent-gap models [27]); their domains of validity and the sensitivity of every headline number to them are quantified in Supplements S5–S6 and S9–S11.

Relation to Ge-hole platforms. The closest precedent is Luttinger–Kohn-based Majorana design for gate-defined Ge hole channels [28]. The silicon case differs in the decisive inputs: a CMOS-native, same-crystal materials stack with a possible all-silicon superconducting parent (Si:B); a smaller split-off gap making six-band treatment mandatory; the [110] FinFET g -tensor phenomenology; and, on the electron side, the conduction-valley step mechanism, which has no Ge analogue and supplies this paper’s sharpest falsifiable prediction.

What would falsify this study. (i) Miscut/step-density-resolved gap spectroscopy on proximitized Si *electron* devices showing induced-gap hardness independent of step density would falsify the valley-step mechanism. (ii) A measured device-geometry [110] hole g -tensor outside $1.4\text{--}2.3$, or strongly gate-dependent, would falsify the six-band design inputs. (iii) Thin-film Si:B with $B_{c\parallel}$ far below the Pauli limit would eliminate the all-silicon detection scenario. (iv) An Al/Si-hole interface with a hard induced gap would, conversely, promote the strongest scenario from hypothetical to testable — the framework is falsifiable in both directions.

Fig 12 — transport gap of the engineered-Si wire ($\alpha=0.05$ eV·Å, $\Delta=50$ μ eV, $B=1.5$ T, $\mu=0$, $L=2$ μ m, $dx=2.5$ nm; normal leads $\mu_{\text{lead}}=2000$ μ eV)

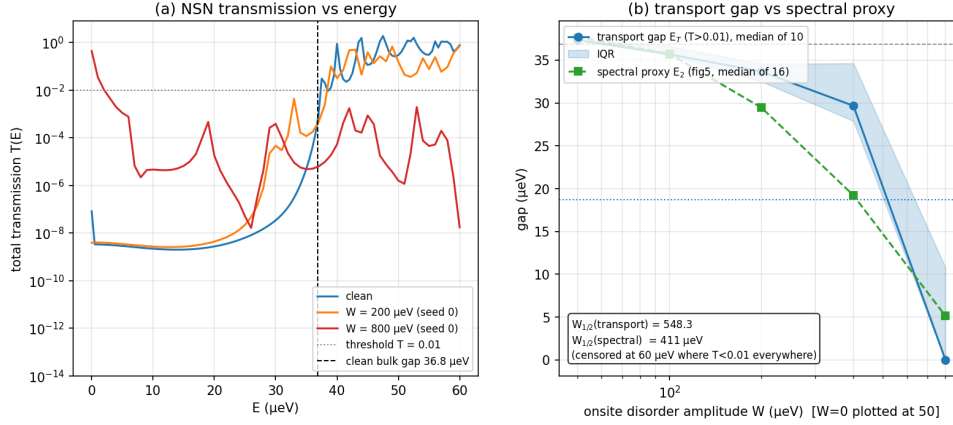


Figure 5: Transport vs spectral disorder thresholds (engineered-electron case): (a) transmission spectra; (b) median transport gap vs disorder with the spectral proxy overlaid.

Data and code availability

All figures and quoted numbers regenerate from the committed code (`run_analysis.py`, `transport.py`, `transport_valley.py`, `lk_holes.py`, `qp_poisoning.py`, `convergence.py`, `tests.py`); the numerical ledger is `output/data/key_numbers.json`, and per-figure regeneration commands, the environment specification, and seed families are documented in `README.md` and Supplement S4/S10. The full repository — code, figures, numerical ledger, scan checkpoints, test suite, review-round corrections log, and this manuscript’s source — is public at

[https:](https://github.com/wtrevena/feasibility-of-creating-majorana-modes-in-silicon-nanowires)

[//github.com/wtrevena/feasibility-of-creating-majorana-modes-in-silicon-nanowires](https://github.com/wtrevena/feasibility-of-creating-majorana-modes-in-silicon-nanowires)

Version synchronization: this manuscript (draft v4), the Supplement (v4), the repository code, and the numerical ledger are a single analysis state: the tagged commit recorded in `MANIFEST.json`, whose curated map ties every headline number in this paper to its ledger key and is verified in continuous integration (`tools/manifest.py --check`). An archival snapshot (tagged commit and DOI) will additionally be deposited on Zenodo at submission. [Zenodo DOI: to be inserted at submission.]

Acknowledgments and provenance

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